

Active Directory Certificate Services (AD CS) Configuration

After completing the core infrastructure setup, Active Directory Certificate Services (AD CS) was installed and configured on Windows Server 2022 (DC01) to provide certificate-based security within the enterprise domain environment.

1. Server Information

Parameter	Value
Server Name	DC01
FQDN	DC01.mahfujidb.com
Domain Name	mahfujidb.com
Role	Domain Controller
Service Installed	Active Directory Certificate Services (AD CS)
CA Type	Enterprise Root CA

Objective

To configure an Enterprise Root Certification Authority (CA) to issue, manage, and validate digital certificates within the Active Directory domain environment.

2. AD CS Installation

AD CS role was installed on the domain controller using Server Manager.

Installation Procedure

1. Open **Server Manager**.
2. Click **Manage** → **Add Roles and Features**.
3. Select **Role-based or feature-based installation**.
4. Select server:

DC01.mahfujidb.com

5. Select **Active Directory Certificate Services**.
6. Click **Add Features**.
7. Continue wizard with default settings.
8. Under **Role Services**, select:

- Certification Authority

9. Click **Install**.

Result

Active Directory Certificate Services role was successfully installed on DC01.

3. AD CS Configuration

After installation, AD CS was configured as an Enterprise Root CA using the configuration wizard.

Configuration Procedure

1. Open **Server Manager**.
2. Click notification flag.
3. Select **Configure Active Directory Certificate Services**.
4. Provide credentials:

MAHFUJIDB\Administrator

5. Select Role Services:
 - Certification Authority
6. Select Setup Type:
 - Enterprise CA
7. Select CA Type:
 - Root CA
8. Select:
 - Create a new private key

5. Cryptography Configuration

Setting	Value
Provider	RSA#Microsoft Software Key Storage Provider
Key Length	2048 bits
Hash Algorithm	SHA256
Private Key	New

Result

A new cryptographic key pair was successfully generated for the CA.

5. CA Name Configuration

The Certification Authority name was configured as:

6. Validity Period Configuration

Parameter	Value
Validity Period	5 Years
Expiration Date	02 June 2031

7. Certificate Database Configuration

Default database locations were used:

Component	Path
Database	C:\Windows\System32\CertLog
Logs	C:\Windows\System32\CertLog

8. Final Configuration Summary

Parameter	Value
CA Type	Enterprise Root CA
Key Type	RSA 2048
Hash Algorithm	SHA256
Setup Type	Enterprise
Private Key	Newly Generated
Status	Successfully Configured

Result

Active Directory Certificate Services was successfully configured as an Enterprise Root Certification Authority on DC01.mahfujidb.com. The CA is now active and capable of issuing and managing certificates within the domain environment.